



Dwight Look College of

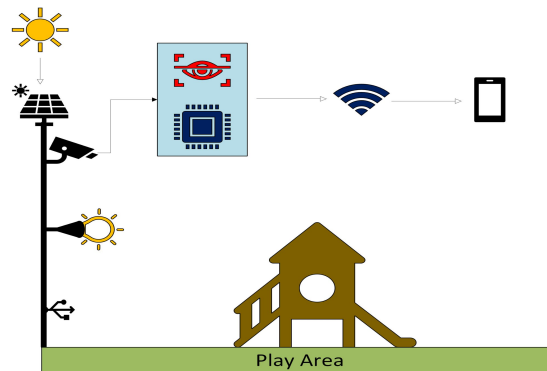
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404 Final Presentation Team 65: Solar Playground with Security System

**Team Members: Zayan Prasla, Alex Allahar, Jackson
Kitchel, Victor Igbokwe
Sponsor: Wonhyeok Jang
TA: Zian Wang**

Problem Overview

A solar-powered security system with a motion-detecting camera. This system provides near real-time video feeds and sends automated alerts for suspicious activity through a mobile app, enhancing safety in these areas. It solves the problem of unreliable surveillance in remote or off-grid locations by offering an autonomous, low-maintenance solution.



Integrated Project Diagram

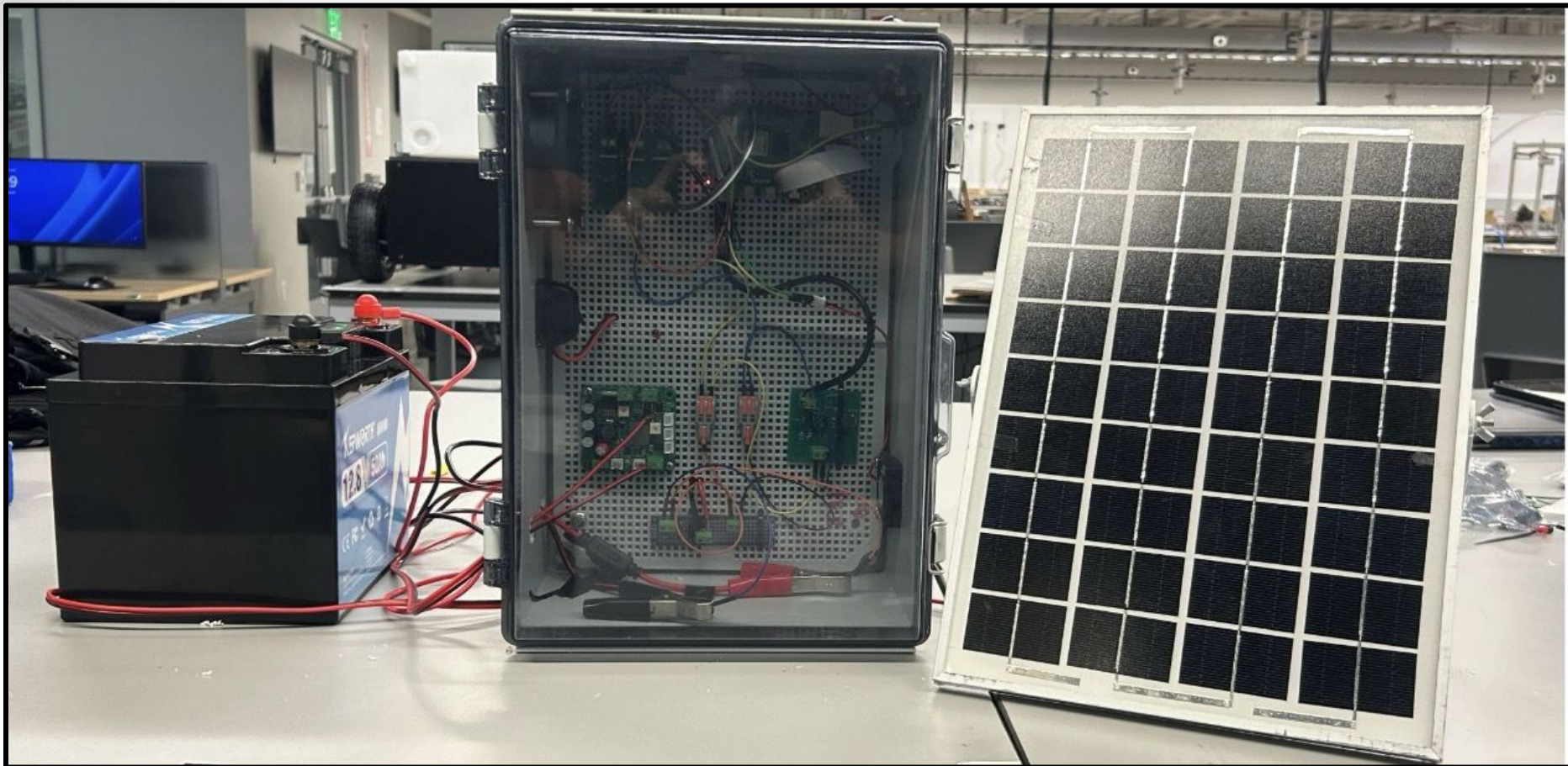


Figure 1: Full System Overview

Integrated Project Diagram (Hardware)

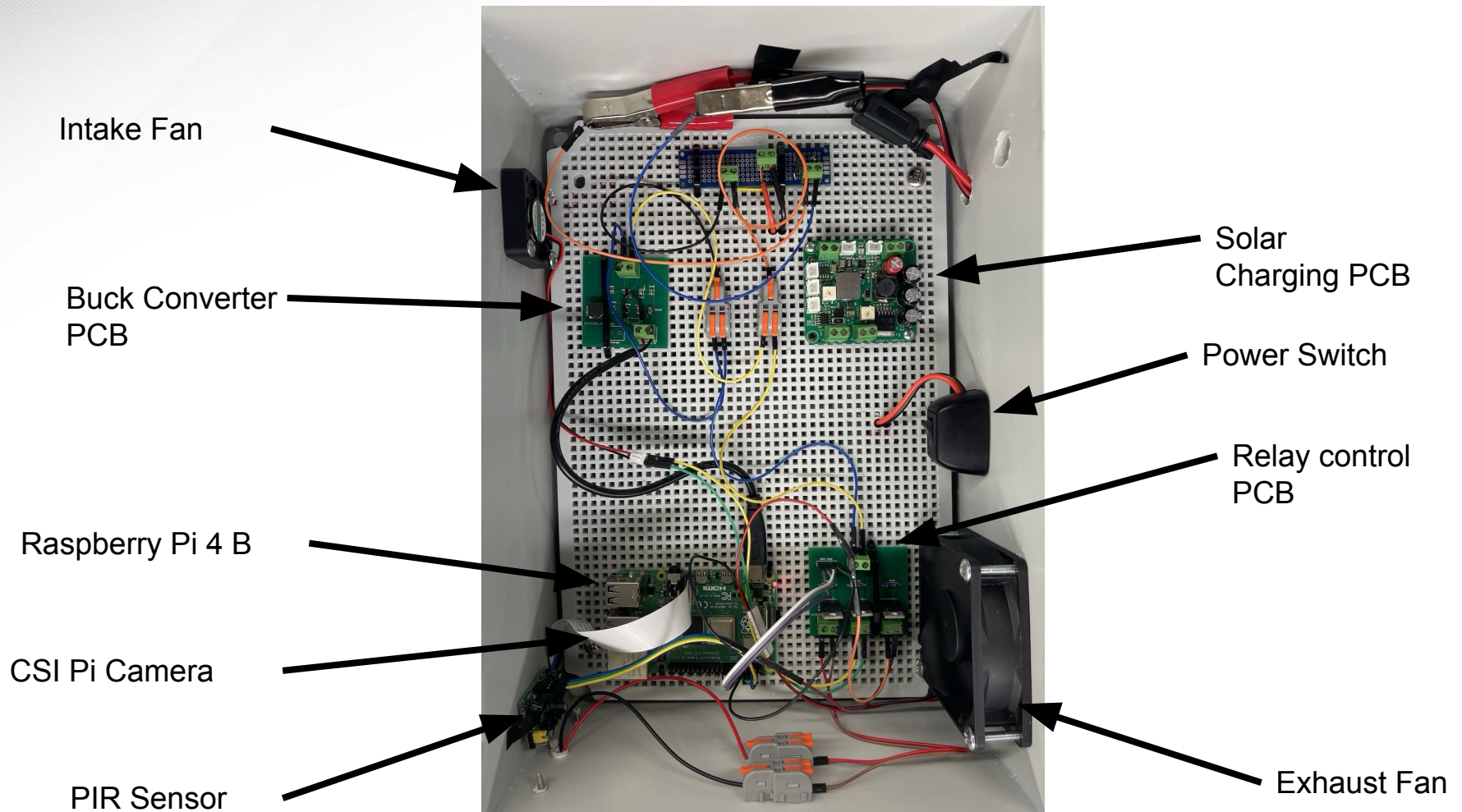


Figure 2: Hardware Overview

Integrated System Diagram (Software)

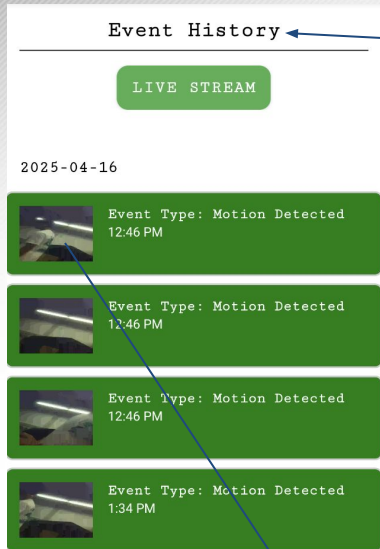


Figure 4:
Event History
Page with
live stream
capability
and
playable
videos

Figure 5: Video
Player from the
app

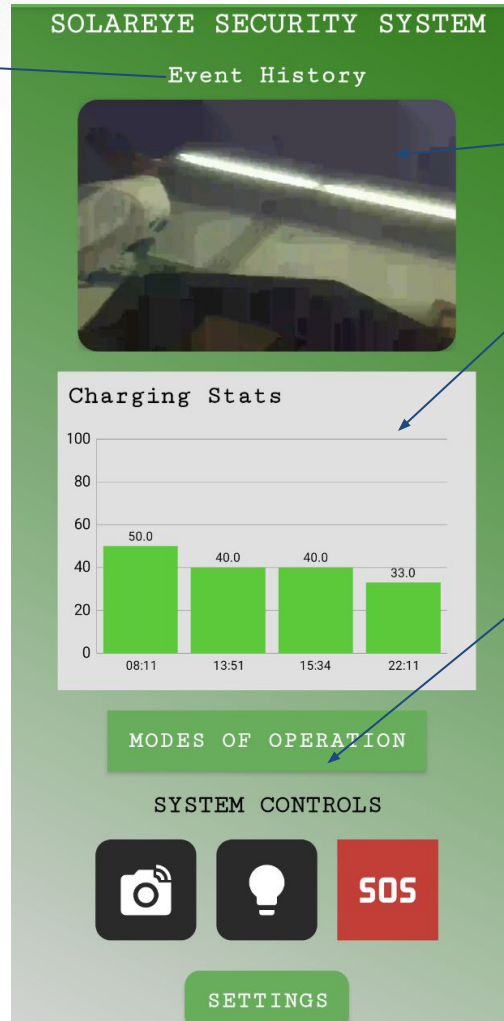
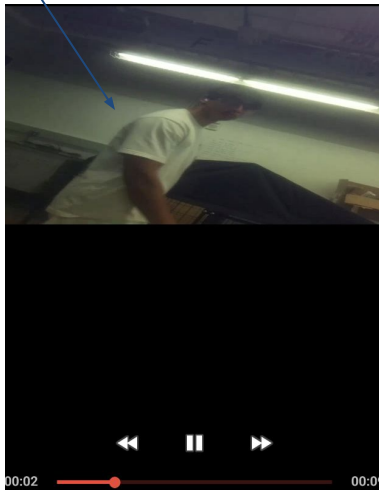
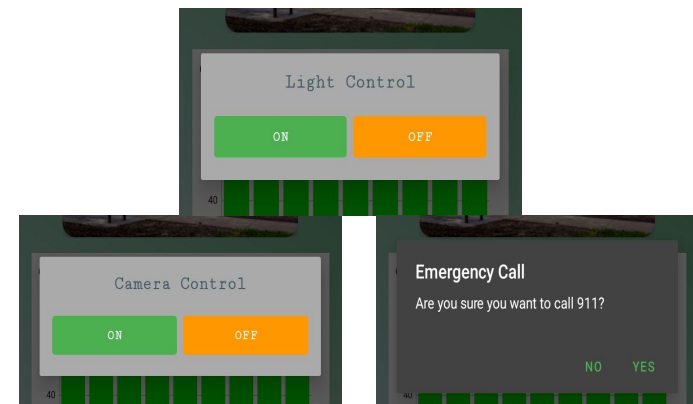
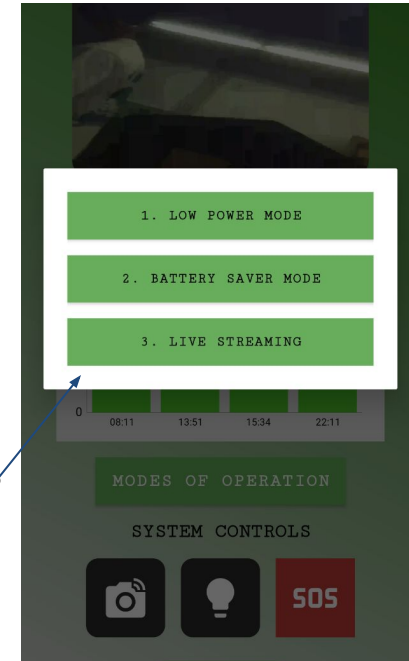


Figure 3:
Homepage with
updating Event
thumbnail and
updating
Charging
Statistics

Figure 6: Modes
of Operation
and System
Controls with
successful
communication





Power Generation - Jackson Kitchel

Designed & Built:

- Designed and built a charging circuit capable of charging a 12V battery using a 100W solar panel.
- Implemented Maximum Power Point Tracking (MPPT) using the BQ24650 IC chip
- Implemented 2 current and voltage sensors using the ACS723 IC chips along with ADC

Tested & Validated:

- Charging voltage test completed along with high/low voltage cutoffs. Sensors tested with MCU. Battery life to be tested.

Challenges & Solutions:

- Redesigned charging circuit to solve incorrect capacitor problem, along with making the circuit more compact.

Power Generation - Jackson Kitchel

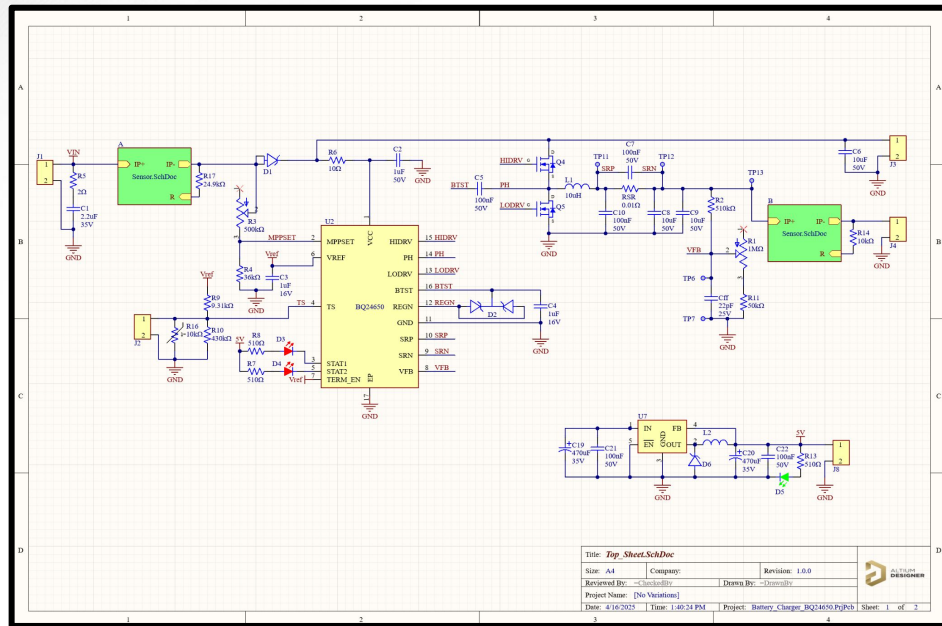


Figure 7: Charging Circuit Schematic

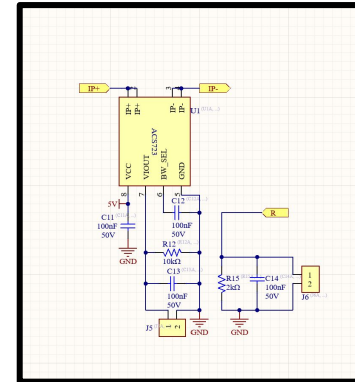


Figure 8: Current Sensor Schematics

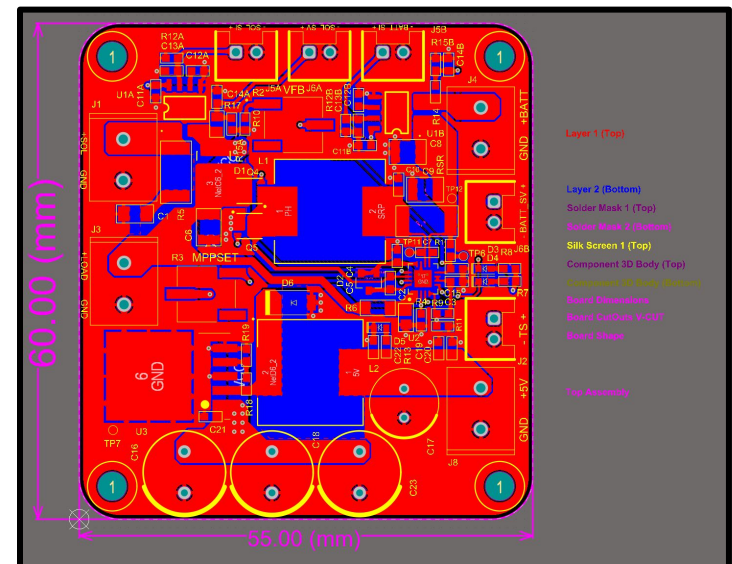


Figure 9: PCB Design

Power Generation - Jackson Kitchel

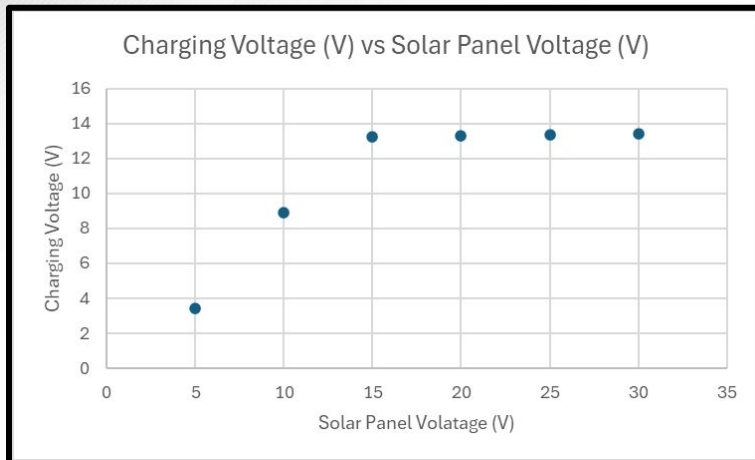


Figure 10: Charging Voltage vs Solar Panel Voltage Graph

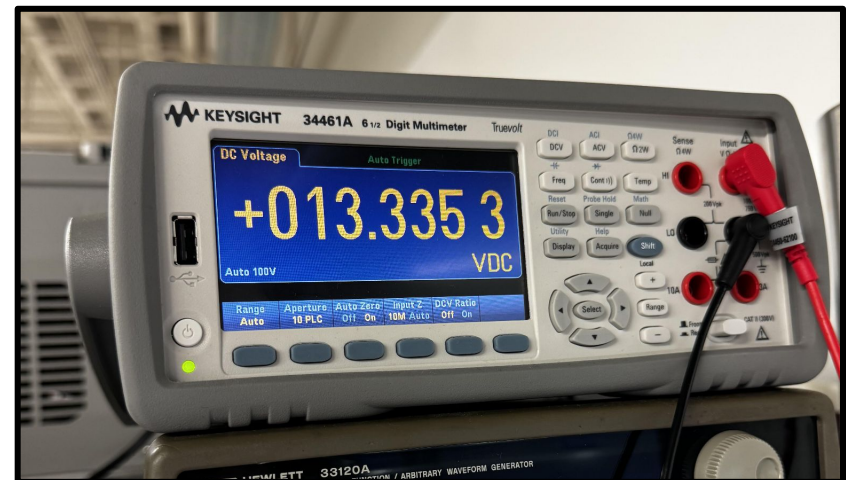


Figure 12: Charging Voltage Output

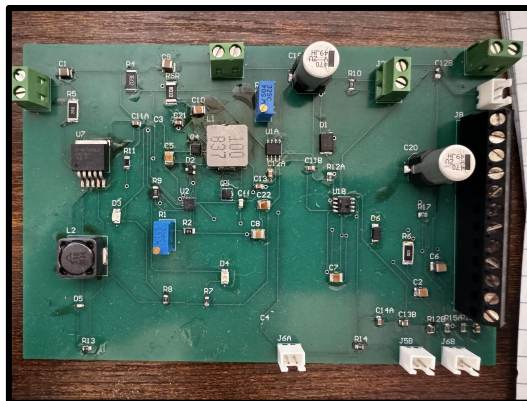


Figure 11: Old PCB Design

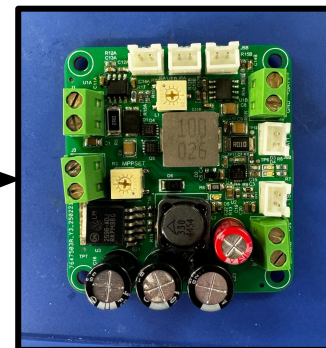


Figure 13: New PCB Design



Power Distribution - Victor Igbokwe

Designed & Built:

- Designed and built a buck converter using the Texas Instruments **TPS563200 DCR** regulator to provide stable 5V output for embedded systems.
- Implemented a relay control module using an **N-channel MOSFET** to act as a switch, enabling the Raspberry Pi to control the on/off state of external loads.

Tested & Validated:

- Buck converter Voltage and load regulation tests, Relay Control module Load regulation test, measured time delay between Pi signal and device activation.

Challenges & Solutions:

- **Safety for repairs:** Installed a switch button to the Junction box to shut down all power flowing through the system
- **Buck Converter:** Re-designed to ensure correct functionality

Power Distribution - Victor Igbokwe

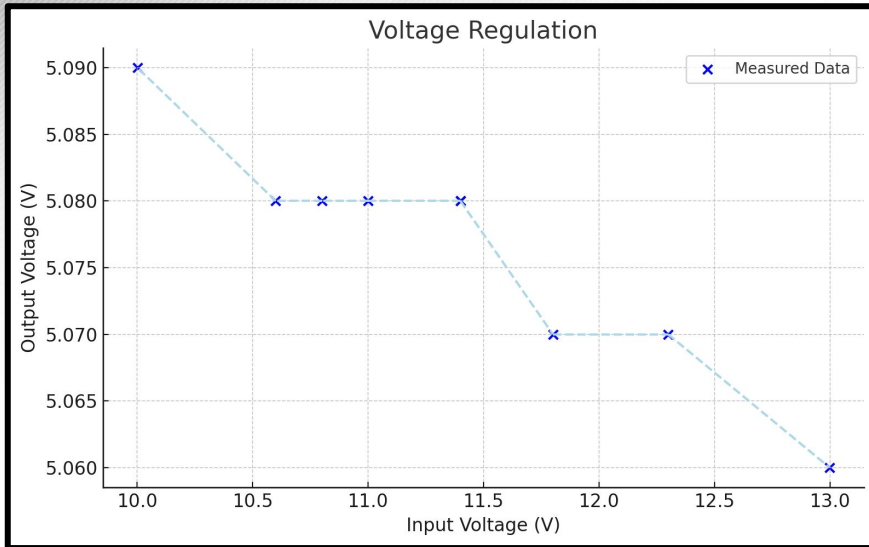


Figure 14: Voltage regulation graph

Time delay powering MPU	1.26 minutes
Raspberry Pi 4B input Voltage Range(V)	4.75 - 5.25 * 5% Tolerance

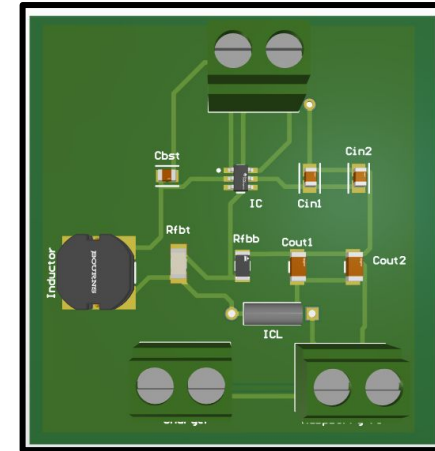


Figure 15: Buck Converter

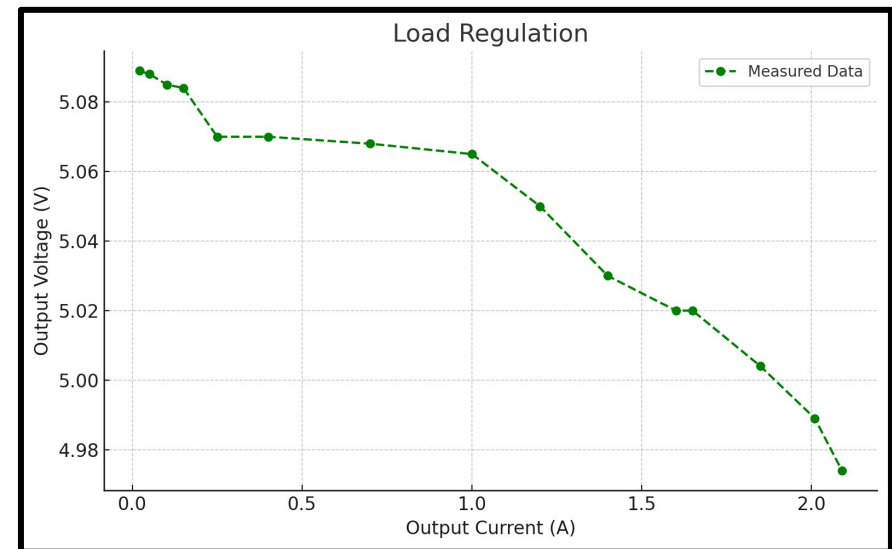


Figure 16: Load regulation graph

Power Distribution - Victor Igbokwe

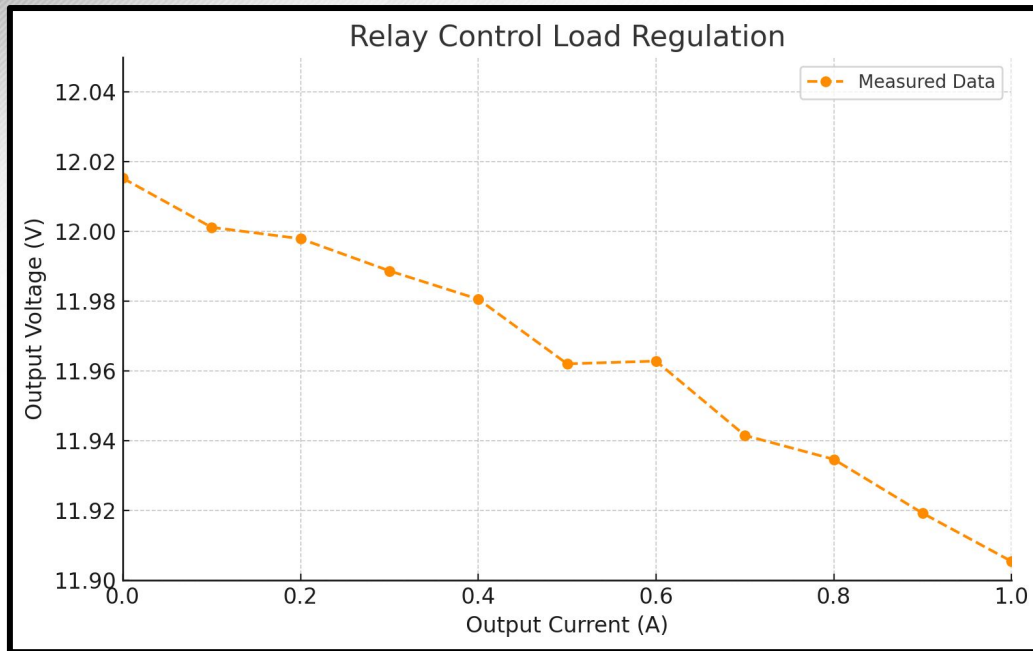


Figure 17: Relay control load regulation graph

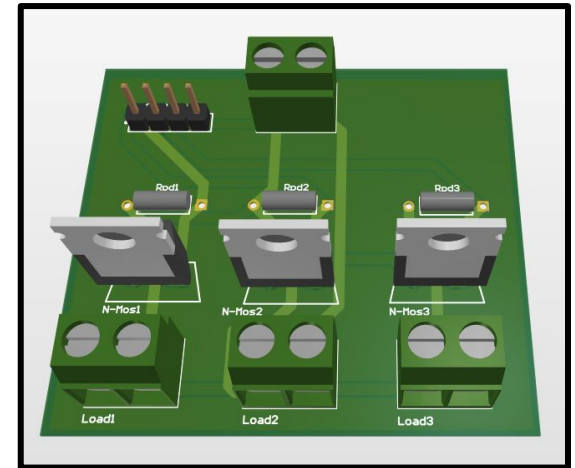


Figure 18: Relay control module PCB

**Time delay between
MPU signal and device
activation**

0.4 seconds

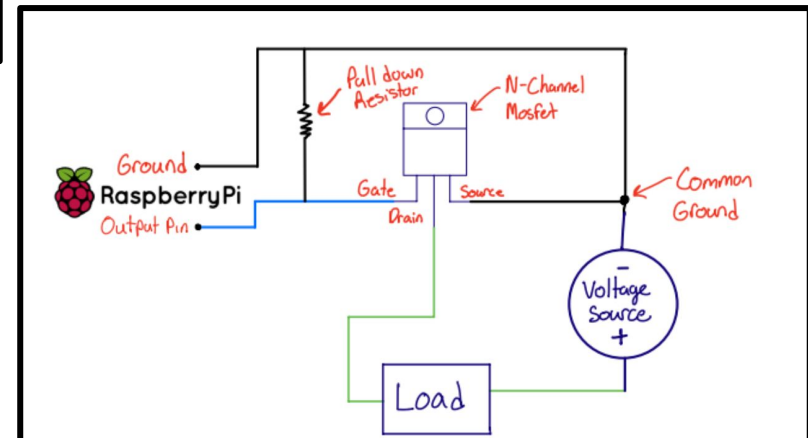


Figure 19: Use of N-Channel Mosfet diagram



Edge Processing Unit - Alex Allahar

Designed & Built:

- Integrated PIR sensor (I2C), voltage sensor (I2C), camera (CSI), LED and fans (GPIO) with a Raspberry Pi 4B to enable the following features:
 - Low-Power Mode
 - Motion Detection
 - Live Streaming
 - LED Deterrence

Tested & Validated:

- Calibrated sensors & fans, ADC communication, Camera quality, power consumption, overall hardware-software latency

Challenges & Solutions:

- **CPU Overload:** Solved with multi-threading and modes of operation controls.
- **MPU Overheating:** Fans for Overheating.
- **Camera:** Replaced Camera (USB to CSI), video format selection, Live streaming delay



Edge Processing Unit - Alex Allahar

Requirements	Target	Measured	Status
Motion to App Delay	< 1 s	0.0784 s	Passed
Video Upload Latency	< 3 s	1.2951 s	Passed
Real Event vs. Video Capture Ending	< 5 s	4.03 s	Passed
Livestream Delay	< 5 s	~7.8 s	Failed
Motion Detection Range	12 ft	16 ft	Passed
Fans Power Consumption	< 5 W	8.8 W	Failed
Pi Power Consumption (Low Power)	< 20 W	9.4 W	Passed
Pi Power Consumption (Default)	< 20 W	10.75 W	Passed
Pi Power Consumption (Livestream)	< 20 W	18.25 W	Passed

Figure 20: Requirements Validation Table



Android Application - Zayan Prasla

Designed & Built:

- Created a secure Firebase Authentication system for account creation, email verification, password reset, password policy and persistent login
- Developed camera, LED, and SOS control features, with Modes of Operation for system power usage
- Implemented real-time event history with video thumbnail previews and video playback

Tested & Validated:

- Verified user authentication flows, password policies, and email verification timeouts
- Confirmed Firestore integration for reading and writing system control states
- Ensured proper display and playback of video thumbnails from Firestore Database URLs

Challenges & Solutions:

- Firestore path issues: Fixed with proper document structure and logging
- Video Player issues: Solved by having the mcu create video url links for the app to open from Firestore Database.

Android Application - Zayan Prasla

App Features Validation Table:

Requirements	Spec	Measured
Authentication Timeout	Email verification link expires in 5 min	Verified expiration at ~5:01 min
Password policy enforcement	Min 6 char, 1 capital, 1 symbol, 1 number	100% rejection rate on invalid attempts
Video event thumbnail loading	Thumbnail shown within 2 sec	Avg load time: 1.2 sec, max: 1.8 sec
Battery graph recency	Only last 24h voltage data shown	Consistently accurate range filter applied
Mode change response (Firestore write)	Update reflected in < 1.5 sec	Avg: 0.9 sec, Min: 0.6 sec, Max: 1.4 s

Figure 21: App Features Validation Table:

Integrated System Results

End-to-End System Operation (Low Power):

Low Power Mode Predicted Run Time $\approx$ 32.8 hours

Lower Power Mode (User Experience):

- Able to see when the event occurred, and duration via but no video.
- LED Deterrence is available, flashes the LED rapidly.



Figure 22: LED Deterrence Active

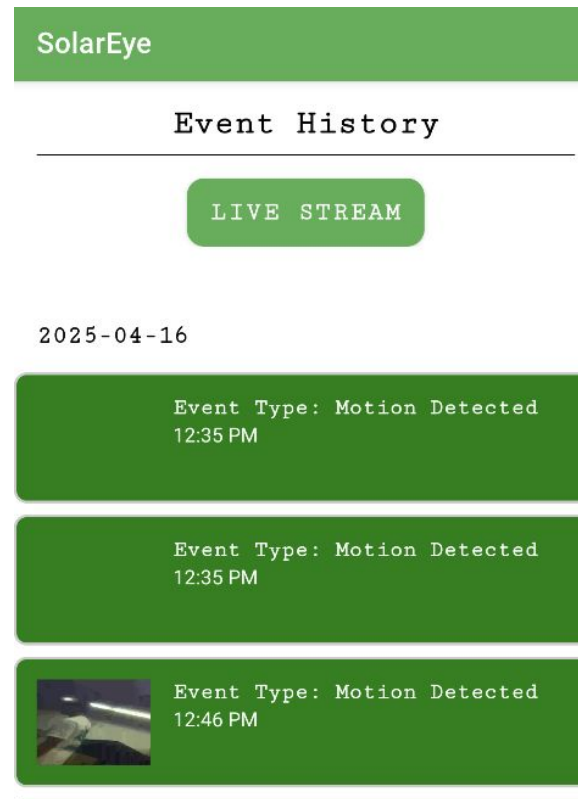


Figure 23: Event History

Integrated System Results

End-to-End System Operation (Default):

Motion Detection Mode Predicted Run Time $\approx$ 30.69 hours

Motion Detection Mode (Pipeline):

- PIR Sensor detects motion $\rightarrow$ event is logged $\rightarrow$ camera starts recording $\rightarrow$ recording stops after 2 seconds of no motion $\rightarrow$ video saved locally(h264) $\rightarrow$ uploaded to Firebase(mp4) $\rightarrow$ app receives video with timestamp $\rightarrow$ user views it in event history.

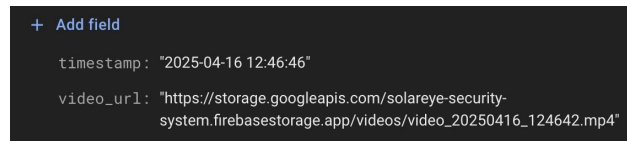


Figure 25: Notification Packaged for App

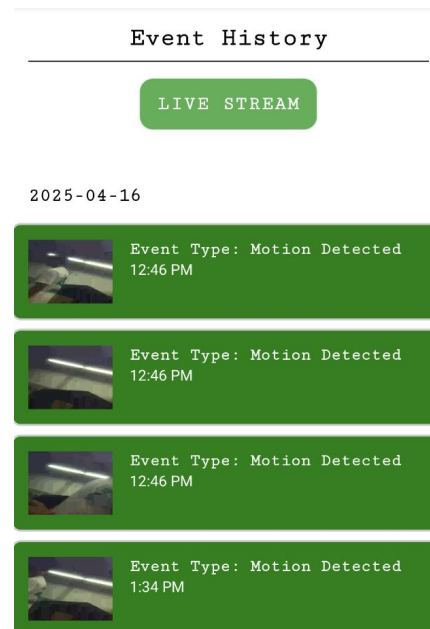


Figure 26: Event History

Figure 24: Motion Detection Capture

Integrated System Results

End-to-End System Operation (Livestream):

Livestream Mode Predicted Run Time $\approx$ 22.18 hours

Lower Power Mode (User Experience):

- User click Livestream Button - HTTP Server Activates - Pi starts HLS Stream Protocol - Stream HTTP Link create/sent to app - user can access stream by clicking on the Link

```
* Serving Flask app "livestream"
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment.
Use a WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8000
* Running on http://10.247.37.101:8000
Press CTRL+C to quit
[STREAM] Livestream started.
10.246.73.49 - - [16/Apr/2025 13:39:57] "GET / HTTP/1.1" 200 -
10.246.73.49 - - [16/Apr/2025 13:39:57] "GET /video/index.m3u8 HTTP/1.1" 404 -
10.246.73.49 - - [16/Apr/2025 13:40:09] "GET / HTTP/1.1" 200 -
10.246.73.49 - - [16/Apr/2025 13:40:10] "GET /video/index.m3u8 HTTP/1.1" 200 -
```

Figure 25: Livestream HTTP Server Startup

+ Add document	+ Start collection
camera_state	+ Add field
livestream >	livestream_url: "http://10.247.37.101:8000"

Figure 26: Livestream HTTP Link to Firebase

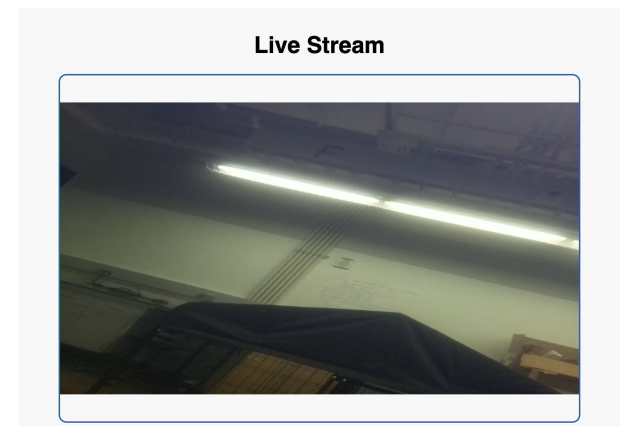


Figure 27: Livestream UI



Conclusions

Changes from original System design:

- Buck Converter IC chip change leading to re-design of Buck converter
- Relay control module PCB added to system design
- Charging Circuit redesigned to solve capacitor problem and consolidate PCB
- ADC added to system to allow communication between current sensor IC chips and MCU

Current Status:

- Integration Complete
- Operational Testing Complete
- Validation Complete
- Edge Case Testing in Progress